Validation and Accuracy Evidence

Kaduna.lk (R26-SE-026) · Geo-Intelligence & Traffic Impact Analysis · Marsook Mohamed Asath (IT22633422) · 21 October 2026

Every number below is printed by `scripts/report_metrics.py` or taken from the named result file. Nothing is quoted from memory.

1. Results at a glance

Test	Measure	Result	Target	Outcome
Impact score vs 120 SUMO simulations	Pearson r / Spearman ρ	0.60 / 0.67	r ≥ 0.75	**Missed**
Lanes-blocked factor alone vs SUMO	Pearson r	0.88	–	Strongest factor
Score error vs simulated severity	MAE (1–10 scale)	3.157	< 1.5	**Missed**
Impact hotspots (DBSCAN)	Clusters · silhouette	25 · 0.64	> 0.3	**Met**
Hotspots vs accident blackspots	Centres within 1.5 km	10 of 25 (≈5 by chance, p ≈ 0.02)	–	**Above chance**
High-impact incidents inside hotspots	Odds ratio	2.3	–	**Above 1**
Hotspot coverage of high-impact incidents	Share inside a hotspot	34.6%	> 80%	**Missed**
Impact-first dispatch (H2)	Network delay vs nearest-first, 0.88 load	19.1% worse, 0 of 30 seeds won	Lower delay	**Rejected**
Forced rerouting (real Galle Rd sub-network)	Network travel time	+18.8 veh-h, worse in 96 of 96	Lower	**No benefit**
Transfer to a real sub-network	Pearson r (full score)	0.20	–	**Weak**
Live scoring API	Automated tests · p95 latency	146 pass · under 10 ms	500 ms	**Met**

> What the evidence supports: an interpretable 1–10 triage score whose ranking follows simulated congestion (ρ 0.67, driven by lanes blocked), 25 impact-weighted hotspots that line up with known blackspots, and a deployed, tested scoring service. What it does not support: ordering the dispatch queue by impact (H2 rejected) and rerouting. The score is therefore used for triage and exploratory unit placement.

2. How each test was run

2.1 Score accuracy against SUMO traffic simulation

SUMO (Simulation of Urban MObility) is the ground truth. `scripts/run_sumo_validation.py` builds 120 road scenarios with a fixed random seed (42). Each one samples road class, lane count, lanes blocked (1–2), demand (up to oversaturation), time of day and all seven incident types independently, so every score factor varies. SUMO simulates each road twice, without and with the blocked lanes, and records the speed drop, queue and vehicle-hours lost. The impact score for the same scenario is then compared with the simulated speed drop. Results are in `data/sumo_results.csv`.

The deployed expert weights give r 0.60 and ρ 0.67, below the 0.75 target. Per factor, lanes blocked explains most of the signal (r 0.88). Weights fitted to SUMO reach r 0.92 on held-out folds (leave-one-road-type-out 0.924), but they are a sensitivity result only and are not deployed, because the simulation grid cannot separate incident type from lanes blocked.

2.2 Impact hotspots

`scripts/run_hotspot_analysis.py` runs DBSCAN (haversine distance, 0.5 km radius, at least 4 incidents per cluster) and KDE over the 500 scored incidents. The incidents are synthetic but placed on the real Colombo OpenStreetMap network, and road class and lanes are read from each incident's GPS point. Quality is checked three ways: the silhouette score (0.64), the distance from each cluster centre to 8 named Colombo accident blackspots (10 of 25 within 1.5 km, against about 5 by chance), and the odds ratio of high-impact incidents falling inside a hotspot (2.3). 71.2% of incidents are noise by design, which is why the 80% coverage target was missed.

2.3 Does impact-first dispatch help? (Hypothesis H2)

`scripts/dispatch_baseline.py` and `scripts/dispatch_load_sweep.py` run a discrete-event dispatch simulation on the real Colombo road network with 15 providers and 30 random seeds. The cost of each incident comes from a model trained on the SUMO results using physical features only, so the score is not graded against itself. At 0.88 fleet utilisation, serving the highest impact score first increased network delay by 19.1% against nearest-first, and won in 0 of 30 seeds. H2 is rejected. Results: `data/analysis/dispatch_load_sweep_summary.json`.

2.4 Engineering checks

The deployed FastAPI service (`rp-group/Main-Repo/components/geo-intelligence`, live at geo.vps.kaduna.lk) has 146 automated tests, all passing on 21 October 2026. Scoring latency p95 is 9.1–9.7 ms against the 500 ms requirement. If the geo service is down, dispatch falls back to a neutral score of 5 after a 2-second timeout, so a help request never fails because of this component.

3. Stakeholder validation

Date	Who	Method	Result
27 Aug 2026	Retired Deputy Inspector General of Police	Interview	Confirmed the problem and a Monday-in, Friday-out traffic pattern. Exposed a bug: every incident was scored as a primary road. Fixed by reading road class from the incident GPS.
5 Oct 2026	Assistant Commissioner, Department of Motor Traffic (20+ years)	Demo, then a written form	SUS 80.0 (above the 68 average). Factor relevance mean 4.4 / 5. A 10-scenario ranking gave ρ −0.05: the expert ranks major roads off-peak higher than the model does.
16 Oct 2026	Automobile Association of Ceylon (call centre and engineering unit)	Demo, then a written form	UMUX-Lite 71.7. Triage diagnosis right in 3 of 3 cases. For brake failure AAC would send a mechanic where the app sends a tow truck.
17 Oct 2026	Team field run, Malabe	Live run with real phones	Full help-request-to-dispatch flow completed end to end on the deployed system.

Each session is one respondent per organisation (n = 1). The results are qualitative evidence, not statistical proof, and none of these organisations uses or endorses the system.

4. Repeating the tests

All commands run from the `RP/` folder with the RP virtual environment's Python. The SUMO 1.26.0 binary installed in that environment must be first on the PATH.

```
export PATH=$PWD/venv/lib/python3.14/site-packages/sumo/bin:$PATH

# Recompute every number in section 1 from the saved results (about 3 s)
venv/bin/python scripts/report_metrics.py

# Re-run the 120 SUMO simulations, seed 42 (about 5 minutes)
venv/bin/python scripts/run_sumo_validation.py

# Geo service test suite (146 tests, about 4 s)
cd ../rp-group/Main-Repo/components/geo-intelligence && .venv/bin/python -m pytest -q
```

Reproducibility check, 21 October 2026: the 120 simulations were re-run from scratch in a separate copy of the project. All 120 rows of `sumo_results.csv` matched the saved results exactly (0 differing values), in 5 min 11 s.

5. Known limits

- The 500 incidents are synthetic. No timed, location-coded incident records exist in Sri Lanka (confirmed verbally by the DMT), so there is no real ground truth to score against.
- The SUMO grid cannot separate incident type from lanes blocked, so the fitted weights are not deployed.
- The congestion predictor ranks scenarios correctly but over-estimates vehicle-hours lost in absolute terms, so its values are reported as uncalibrated.
- The DMT survey item on "detour suggestions" refers to a feature the product does not have. It is reported as a rating of the hotspot map.
- All stakeholder sessions are n = 1.

6. Evidence files

Evidence	File
SUMO scenarios and results	`RP/data/sumo_results.csv`
Hotspot results	`RP/data/hotspot_results.csv`
H2 dispatch results	`RP/data/analysis/dispatch_load_sweep_summary.json`, `RP/docs/so6-results.md`
DMT form and summary	`RP/cursorgenerateddocs/pp2-validation/police-meeting-2026-10-05/`
AAC forms, scans and findings	`RP/cursorgenerateddocs/pp2-validation/aac-visit-2026-10-16/`
Malabe field run log	`RP/cursorgenerateddocs/pp2-validation/field-validation-malabe/live/field-run-2026-10-17.log`
Source code	github.com/JanukshanS/R26-SE-026 (`components/geo-intelligence`)